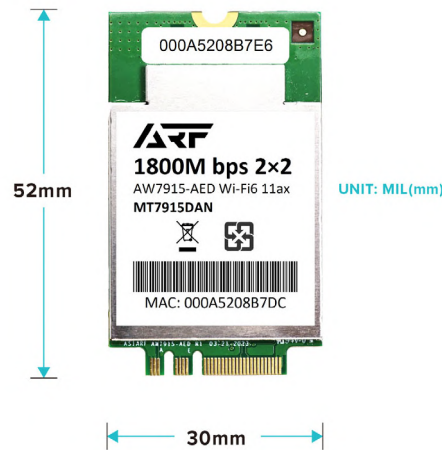


DATASHEET**WiFi 6 11ax M.2 AE Key Module AW7915-AED**

Description

The AW7915-AED is a WiFi6 M.2 AE key Module based on the MediaTek MT7915DAN chipset. It supports a PHY rate of 573+1201 Mbps and complies with IEEE 802.11ax/ac standards. It offers feature-rich wireless connectivity in Dual Bands Dual Concurrent (DBDC) mode, supporting simultaneous operation in the 2.4GHz and 5GHz bands.

The module features optimized RF architecture and baseband algorithms, providing superb performance and low power consumption. Its intelligent MAC design offloads Wi-Fi tasks from the host processor. The MT7915DAN chipset supports standard-based features for security, quality of service, and international regulations.

Deep sleep mode is supported with multiple power domains implemented on the chip, enabling low power consumption. The module includes two CPU systems with 32-bit RISC MCU subsystems for clock control, power management, and host interface configuration. PDMA engines support on-the-fly data buffer management.

The MT7915DAN chipset combines the WiFi MAC and BBP subsystems, working together with the MT7975DN, to deliver best-in-class radio performance and low power consumption. It also supports MU-MIMO with different configurations for enhanced multi-user connectivity.

The module utilizes PCIe2.1 for stable bandwidth between the host platform and the chipset. It provides high data throughput over WiFi and is designed for optimal performance and power efficiency.

In summary, the AW7915-AED M.2 AE key Module offers reliable WiFi6 connectivity, optimized performance, and power efficiency in a compact form factor.

Version: 1.0

Release date: 2023-08-17

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Features

- 2T2R in 2.4G+2T2R in 5G with support of up to 573+1201Mbps PHY rate
- Supports 20, 40, 80 channels
- HE MCS0-11 BW20/40/80MHz with Nss=1~2
- Short Guard Interval
- Space-time block code (STBC)
- Low Density Parity check (LDPC)
- Support digital pre-distortion to enhance PA performance
- Smoothing (channel estimation) extension to MIMO case
- DFS radar detection
- Embedded ARM Cortex R4 processor for full host CPU offload
- Embedded 32-bit RISC microprocessor
- Support STBC, LDPC, TX Beamformer and RX Beamformer
- Decoded BW20/40/80 up to 4x2 MU MIMO feedback
- Greenfield, mixed mode, legacy modes support
- Highly integrated RF with 40nm low power process
- Noise mitigation:
 - Supports background scan function for fast channel switching
 - Supports spectrum analysis for non-Wi-Fi signals
- Intelligent power saving
- WPA WMM, WMM PS (QoS)
- Integrate high efficiency internal 2.4G/5G PAs
- Intelligent Calibration (iCal) reduces the production time

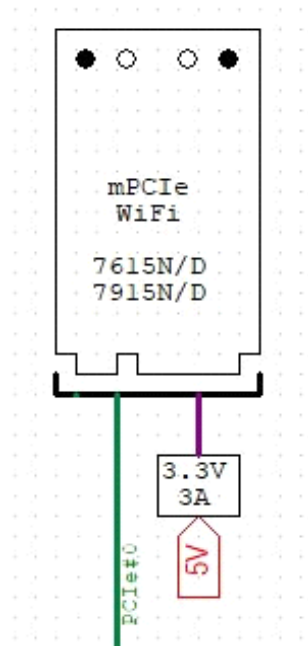


Specifications

Standard	Wireless: IEEE 802.11ax/ac/b/g/n (2Tx2R)
BUS Interface	M.2 AE Key
Data Rate	IEEE 802.11ax 1201Mbps@TX/RX IEEE 802.11ac 1773Mbps@TX/RX IEEE 802.11a/n 600Mbps@TX/RX IEEE 802.11a/g 108Mbps IEEE 802.11b 22Mbps IEEE 802.11ac up to 1733Mbps
Output Power	11b: 23dbm+/- 1.5dbm @ 11Mbps 11g: 20dbm+/- 1.5dbm @ 54Mbps 11g/n: 20dBm +/- 1.5dbm @MCS7,HT20, 17dBm@MCS7,HT40 11a: 19.5dBm +/- 1.5dbm @ 54Mbps 11a/n: 19.5dBm+/- 1.5dbm @MCS7,HT20, 17dBm@MCS7,HT40 11ac HT20: 20+/-1.5dBm@MCS8 11ac HT40: 17+/-1.5dBm@MCS9 11ac HT80: 14.5+/-1.5dBm@MCS9 11ax HT20: 20+/-1.5dBm@MCS9 11ax HT40: 17+/-1.5dBm@MCS9 11ax HT80: 14.5+/-1.5dBm@MCS11
Receiver Sensitivity	11b: -99dBm@11Mbps 11g: -95dBm@54Mbps 11g/n: -90dBm@HT20,MCS7, -86dBm@HT40,MCS7 11a: -90Bm@54Mbps 11a/n: -85dBm@HT20,MCS7, -81dBm@HT40,MCS7 11ac: -90dBm+/-2dBm @VHT20 MCS8 11ac: -85dBm+/-2dBm @VHT40 MCS9 11ac: -68dBm+/-2dBm @VHT80 MCS9 11ax: -61dBm+/-2dBm @HE20 MCS11 11ax: -58dBm+/-2dBm @HE40 MCS11 11ax: -55dBm+/-2dBm @HE80 MCS11
Antenna	External Antenna connector (IPEX) x2
Frequency Range	US, Canada 2.412GHz~2.462 GHz (CH1~CH11) EU, Japan: 2.412GHz~2.472GHz (CH1~CH13) 5.15GHz ~5.875GHz
Software	Security: 64/128-bit WEP Encryption, WPA, WPA2 Driver: Linux
Operating Voltage	DC 3.3V ± 5%
Temperature	Operating: 0°C ~ +70°C Storage : -20°C ~ +90°C
Humidity	Operating Humidity: 10%~90% non-condensing Storage Humidity: 10%~90% non-condensing
Dimension	50.95(H) x 30(W)mm
Weight	0.0125 kg



Power consumption maximum is 9.1W, average is 7W. Main board Power Supply design please provide 3.3V 3.5A, minimum 3.3V 3A.

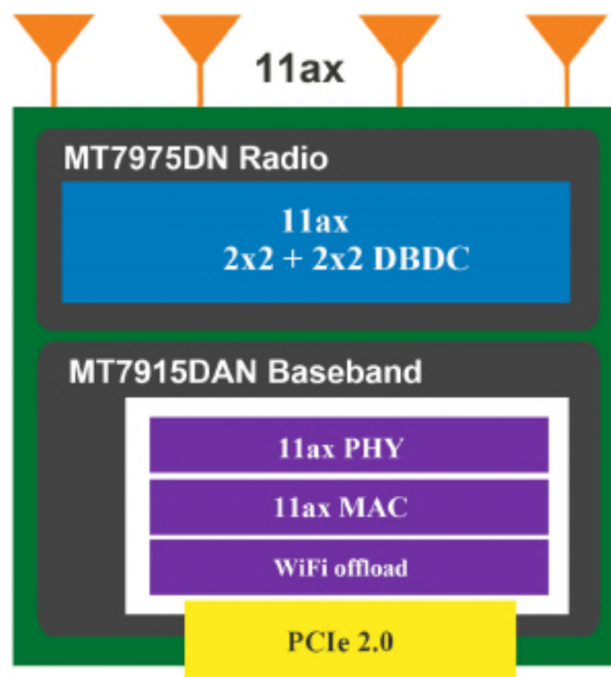


11ax related reading:

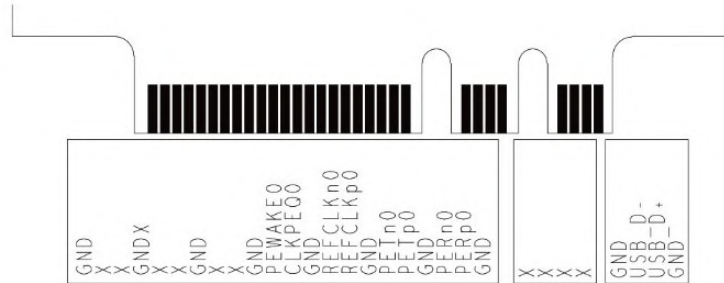
[Introduction to 802.11ax High-Efficiency Wireless - NI](#)

[IEEE 802.11ax - Wikipedia](#)

AW7915-AED Block Diagram



Top side



Bottom side

